

STUDYGENIE-AI

Descriptive Questions & Answers

Subject: Generative AI

1. Study Notes

Generative AI learns patterns from existing data and produces new content that follows those patterns.

Concept overview



Important Concepts

- Generative AI creates new content from learned data patterns.
- GANs use a Generator and a Discriminator.
- Diffusion models generate content through iterative denoising.
- Transformers use attention to model relationships between tokens.

Important Points

- The Generator creates candidate samples.
- The Discriminator evaluates whether a sample is real or generated.
- Model quality depends on training data and evaluation.

Revision Tips

- Remember the Generator creates and the Discriminator evaluates.
- Compare the training and generation process of major model families.

2. Questions & Answers

Q01

What is a GAN?

ANSWER

A Generative Adversarial Network (GAN) is a model made of a Generator and a Discriminator that learn through competition.

Source: Uploaded Study Material, p. 1

Q02

How do diffusion models work?

ANSWER

Diffusion models learn to reverse a noise-adding process and generate data through iterative denoising.

Source: Uploaded Study Material, p. 2